

# SOC Home Lab

## Alert Triage & Incident Report

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|-------------------|--------------------------------------------------------------|
| Analyst           | Parvin S Chandran                                            |
| Environment       | Home Lab (VirtualBox - Kali Linux & Ubuntu Server 24.04 LTS) |
| SIEM Platform     | Splunk Enterprise + Universal Forwarder                      |
| Report Date       | July 2026                                                    |
| Incidents Covered | 3 (SSH Brute Force, Port Scan, Privilege Escalation)         |

### Executive Summary

This report documents three simulated attack scenarios executed against a home-lab Ubuntu victim machine from a Kali Linux attacker, and the corresponding detection logic built in Splunk to identify each one. The lab environment was intentionally segmented (VirtualBox NAT Network for attacker/victim communication, plus a Host-Only adapter for log forwarding to a Windows-hosted Splunk instance) to safely reproduce realistic attacker behavior end-to-end: attack execution, log generation, log forwarding, detection query development, alert configuration, and triage documentation.

## Incident 1 — SSH Brute Force Attack

|                           |                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alert Name</b>         | SSH Brute Force Detected                                                                                                                                                                                                                                                                                                                         |
| <b>Attack Tool</b>        | Hydra (dictionary-based SSH credential attack)                                                                                                                                                                                                                                                                                                   |
| <b>Source IP</b>          | 10.0.2.13 (Kali - Attacker)                                                                                                                                                                                                                                                                                                                      |
| <b>Target</b>             | 10.0.2.15 (Ubuntu - Victim), user: ubuntu                                                                                                                                                                                                                                                                                                        |
| <b>Log Source</b>         | /var/log/auth.log (sourcetype=auth)                                                                                                                                                                                                                                                                                                              |
| <b>Severity</b>           | High                                                                                                                                                                                                                                                                                                                                             |
| <b>Classification</b>     | True Positive (Simulated)                                                                                                                                                                                                                                                                                                                        |
| <b>Analysis</b>           | Multiple failed SSH login attempts were observed from a single source IP within a short window, exceeding SSH's built-in maximum authentication attempt threshold. This pattern - many rapid failures against one account from one IP - is consistent with automated brute-force tooling such as Hydra or Medusa, rather than normal user error. |
| <b>Recommended Action</b> | Block source IP at the firewall/security group level; enforce account lockout policy after N failed attempts; disable password authentication in favor of SSH key-based auth; deploy fail2ban for automated IP banning.                                                                                                                          |

### Splunk Detection Query

```
index=* host="ubuntu-victim" "Failed password"
| rex "from (?\\d+\\.\\d+\\.\\d+\\.\\d+)"
| stats count by src_ip
| where count > 5
```

## Incident 2 — Port Scan (Reconnaissance)

|                       |                                              |
|-----------------------|----------------------------------------------|
| <b>Alert Name</b>     | Port Scan Detected                           |
| <b>Attack Tool</b>    | Nmap (service/version detection scan, -sV)   |
| <b>Source IP</b>      | 10.0.2.13 (Kali - Attacker)                  |
| <b>Target</b>         | 10.0.2.15 (Ubuntu - Victim)                  |
| <b>Log Source</b>     | /var/log/syslog - UFW firewall block entries |
| <b>Severity</b>       | Medium                                       |
| <b>Classification</b> | True Positive (Simulated)                    |

|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analysis</b>           | The source IP generated connection attempts against 9 distinct destination ports within a short window, all blocked by UFW's default-deny policy. Touching many distinct ports in rapid succession is a strong indicator of active network reconnaissance / port scanning, typically the phase preceding an exploitation attempt against a discovered open service. |
| <b>Recommended Action</b> | Block or rate-limit the source IP; verify UFW/firewall default-deny rules remain enforced on all interfaces; monitor for follow-up exploitation attempts targeting any ports found open during the scan.                                                                                                                                                            |

### Splunk Detection Query

```
index=* host="ubuntu-victim" "UFW BLOCK"
| rex "SRC=(?\d+\.\d+\.\d+\.\d+)"
| rex "DPT=(?\d+)"
| stats dc(dest_port) as unique_ports_hit, count as total_attempts by src_ip
| where unique_ports_hit > 5
```

## Incident 3 — Privilege Escalation Attempt

|                           |                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alert Name</b>         | Privilege Escalation Attempt Detected                                                                                                                                                                                                                                                                                                                                            |
| <b>Attack Method</b>      | Repeated failed sudo and su authentication attempts                                                                                                                                                                                                                                                                                                                              |
| <b>Source</b>             | Local session, user: ubuntu, target: root                                                                                                                                                                                                                                                                                                                                        |
| <b>Target</b>             | 10.0.2.15 (Ubuntu - Victim)                                                                                                                                                                                                                                                                                                                                                      |
| <b>Log Source</b>         | /var/log/auth.log - PAM authentication failure events                                                                                                                                                                                                                                                                                                                            |
| <b>Severity</b>           | High                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Classification</b>     | True Positive (Simulated)                                                                                                                                                                                                                                                                                                                                                        |
| <b>Analysis</b>           | Multiple failed authentication attempts were logged via both sudo and su, all targeting the root account. Repeated failures specifically aimed at obtaining root - rather than isolated typos - are a strong indicator of an account being probed for privilege escalation, whether by an external attacker who has already gained low-privilege access, or a malicious insider. |
| <b>Recommended Action</b> | Investigate the source session/user for signs of compromise; enforce strong password and MFA policy for privileged access; consider restricting su/sudo to a defined admin group; alert immediately on any successful escalation following repeated failures.                                                                                                                    |

### Splunk Detection Query

```
index=* host="ubuntu-victim" ("pam_unix(sudo:auth): authentication failure" OR "pam_unix(su:auth): authentication failure")
| rex "logname=(?\S+)"
| rex "user=(?\S+)"
| eval escalation_type=if(match(_raw, "sudo:auth"), "sudo", "su")
| stats count as failed_attempts, values(escalation_type) as methods_used by user, target_user
```

```
| where failed_attempts > 2
```

## Conclusion

All three simulated attacks were successfully detected using custom Splunk Search Processing Language (SPL) queries built from raw Linux log sources, with corresponding alerts configured to notify on matching activity. This exercise demonstrates a full detection engineering workflow: identifying attacker behavior signatures in raw logs, building targeted correlation searches, tuning detection thresholds against realistic traffic volumes, and documenting findings in a triage-ready format suitable for SOC analyst review.